

A Margolus–Levitin speed limit for observables: mean energy bounds expectation-value change quadratically

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The Mandelstam–Tamm and Margolus–Levitin quantum speed limits bound how fast a *state* evolves, using the energy variance and the mean energy above the ground state, respectively. Speed limits on *observables*—the change of an expectation value $\langle A(t) \rangle$ —have so far used the energy *variance* (the Mandelstam–Tamm/quantum-Fisher-information lineage); the *mean* energy has not been brought to bear on the change $\Delta \langle A \rangle$ in a fixed state. We close this branch. First, we prove a *no-go* theorem: there is no state-independent *linear* mean-energy bound on the time to change an observable’s expectation by $\Delta = |\langle A(T) \rangle - \langle A(0) \rangle|$; the optimal *state-independent* mean-energy exponent of Δ is exactly two. Second, we prove the corresponding sharp *quadratic* bound, $T(\langle H \rangle - E_0) \geq C_\star \Delta^2 / \sigma_A^2$, valid for every time-independent Hamiltonian H (ground energy E_0), every bounded observable A with spectral spread $\sigma_A = (\lambda_{\max} - \lambda_{\min})/2$, and every pure or *mixed* state, with the dimension-independent constant $C_\star = 1/(8 \sin x_\star) = 0.172506267461\dots$, where $x_\star$ is the smallest positive root of $\tan(x/2) = x$. The bound is tight, saturated by a near-ground two-level family. We then promote it to the *exact* energy–time/swing trade-off curve of which $C_\star$ is the small-swing slope—tight at every swing, the observable analog of the Giovannetti–Lloyd–Maccone curve for states—prove the full constant survives for mixed states via joint convexity of the trace distance, sharpen the constant for bandwidth-limited generators and for several observables at once, and show the quadratic law degrades to a linear one when the initial state is an eigenvector of the observable (a sufficient condition). The result completes the previously empty (mean-energy $\times$ observable) corner of the speed-limit landscape and, being quadratic, is most constraining where the linear variance bounds are weakest. We discuss its cleanest physical home—autonomous quantum clocks, where it furnishes a coherent mean-energy resolution floor complementary to the known entropy and rate bounds—and delimit honestly the settings it does not constrain.

I. INTRODUCTION

Quantum speed limits (QSLs) quantify how fast a quantum system can change. Two foundational results anchor the field. The Mandelstam–Tamm (MT) bound [1] uses the energy *variance* $\Delta H = \sqrt{\langle H^2 \rangle - \langle H \rangle^2}$: the time to reach an orthogonal state obeys $T_\perp \geq \pi/(2\Delta H)$ ($\hbar = 1$). The Margolus–Levitin (ML) bound [2] uses instead the *mean* energy above the ground state, $T_\perp \geq \pi/[2(\langle H \rangle - E_0)]$; the two are unified and tight together [3]. Both, and the large literature that followed [4], bound the motion of the *state* (fidelity, Bures angle, orthogonalization).

A parallel and rapidly growing line bounds the motion of an *observable*—the rate of change of an expectation value $\langle A(t) \rangle = \text{Tr}[\rho(t)A]$ —which is what an experiment actually reads [5–8]. This program now includes a first-of-kind multiparticle experiment [9]. With few exceptions, these observable QSLs bound the expectation-value *rate* through the energy *variance*. The unifying bound of García-Pintos *et al.* [5] reads $|\dot{a}| \leq \Delta A \sqrt{I_F}$ with I_F the quantum Fisher information ($I_F = 4\Delta H^2$ for pure states); Mohan and Pati [6] obtain $|\text{d}\langle O \rangle/\text{d}t| \leq 2\Delta O \Delta H$; the geometric generalization of Bringewatt *et al.* [7] again uses the (generalized) Fisher information. All are *linear* in the observable change and powered by an en-

ergy *variance*. The *mean-energy* resource that drives the Margolus–Levitin bound has so far reached observables only as an *operator-distance* bound for operator flows [8] or through the mean of a transition Hamiltonian for macroscopic transitions [10]—never as a bound on the change $\Delta \langle A \rangle$ of a bounded observable in a fixed state. This is the gap we fill.

We prove that the mean energy controls observable change *quadratically* and not linearly. Concretely (Sec. IV), for a time-independent H with $E_0 = \min \text{spec}(H)$, a bounded Hermitian A with spectral spread $\sigma_A = (\lambda_{\max}(A) - \lambda_{\min}(A))/2$, and $\Delta = |\langle A(T) \rangle - \langle A(0) \rangle|$,

$$T(\langle H \rangle - E_0) \geq C_\star \frac{\Delta^2}{\sigma_A^2}, \quad C_\star = \frac{1}{8 \sin x_\star}, \quad (1)$$

with $x_\star = 2.331122370\dots$ the smallest positive root of $\tan(x/2) = x$, giving $C_\star = 0.172506267461\dots$. A matching no-go (Sec. III) shows no *linear* mean-energy bound can exist; the quadratic law is the best possible. We then sharpen Eq. (1) to the *exact* tight trade-off curve valid at every swing (Sec. V). We work in units $\hbar = 1$ and, without loss of generality, shift $E_0 = 0$ so that $H \geq 0$; $\hbar$ is restored in the applications.

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II. THE SPEED-LIMIT DICHOTOMY

The result completes a simple 2×2 classification (Table I), organized by *what moves* (a state or an observable) and *which energy functional powers the bound* (the variance or the mean energy).

Three corners were known; the lower-right (mean energy $\times$ observable) was empty, and it is the *only* quadratic one. The mechanism is transparent and is the heart of the no-go. The mean energy bounds only how far the *state* moves—a geometric quantity—through the ML inequality $1 - F \leq K T(\langle H \rangle - E_0)$, $F = |\langle \psi(0) | \psi(T) \rangle|$. An observable change reads that geometry only through $\Delta \leq 2\sigma_A \sqrt{1 - F^2}$, a *square root*. Composing a square root (geometric) with a linear (energetic) estimate yields $\Delta \propto \sqrt{T(\langle H \rangle - E_0)}$, i.e. a quadratic law. By contrast MT bounds the observable *velocity* $|d\langle A \rangle/dt| = |\langle [H, A] \rangle| \leq 2\Delta H \sigma_A$ directly—no square root—hence linear. (Here $K = \sup_{x>0} (1 - \cos x)/x = \sin x_*$; see Sec. IV.)

III. NO-GO: NO LINEAR MEAN-ENERGY BOUND

Theorem 1 (No-go). *For time-independent H and bounded Hermitian A there is no positive state-independent constant κ with $T(\langle H \rangle - E_0) \geq \kappa \Delta / \sigma_A$ for all (H, A, ψ) . More strongly, the sharp mean-energy bound is quadratic: the exponent of Δ is exactly 2.*

Proof. By Eq. (1), $P := T(\langle H \rangle - E_0) \geq C_* \Delta^2 / \sigma_A^2$, and the near-ground two-level family of Sec. V saturates it, $P_{\min}(\Delta) = C_* \Delta^2 / \sigma_A^2$ as an infimum. Hence $P_{\min}(\Delta) / (\Delta / \sigma_A) = C_* \Delta / \sigma_A \rightarrow 0$ as $\Delta \rightarrow 0$: any fixed positive κ is violated by near-ground two-level states. The tight exponent is 2. $\square$

The microscopic reason is that $\langle A(t) \rangle = \sum_{m,n} \bar{c}_m c_n A_{mn} e^{i(E_m - E_n)t}$ is a *signed* sum over Bohr frequencies weighted by coherences, in contrast to the probability-weighted, ground-referenced sum $\chi(T) = \sum_n p_n e^{-iE_n T}$ that the mean energy controls. The coherence weight that sets Δ and the populations that set $\langle H \rangle - E_0$ are independent, which is exactly why a naive linear transfer of the ML state bound fails.

IV. THE SHARP QUADRATIC BOUND

Theorem 2 (Main). *Let H be a time-independent Hermitian operator on a finite-dimensional Hilbert space with $E_0 = 0$ ($H \geq 0$), and let A be a bounded Hermitian observable with spectral spread σ_A . For any pure state, with $\Delta = |\langle A(T) \rangle - \langle A(0) \rangle|$,*

$$T \langle H \rangle \geq C_* \frac{\Delta^2}{\sigma_A^2}, \quad C_* = \frac{1}{8K}, \quad K = \sup_{x>0} \frac{1 - \cos x}{x}. \quad (2)$$

Proof. Work in the energy eigenbasis, $|\psi\rangle = \sum_n c_n |n\rangle$, $p_n = |c_n|^2$. Both sides scale as σ_A^2 , and Δ and the dynamics of $\langle A(t) \rangle$ are invariant under $A \rightarrow A + cI$ (since $\text{Tr}[\rho_T - \rho_0] = 0$); center A so that $\|A\|_{\text{op}} = \sigma_A$ and set $\sigma_A = 1$.

Step 1 (geometry). With $\rho_t = |\psi(t)\rangle\langle\psi(t)|$ and $X = \rho_T - \rho_0$ (traceless, Hermitian), Hölder's inequality gives $\Delta = |\text{Tr}[AX]| \leq \|A\|_{\text{op}} \|X\|_1 = \|X\|_1$. For two pure states the nonzero eigenvalues of X are $\pm\sqrt{1 - F^2}$, $F = |\langle \psi(0) | \psi(T) \rangle|$, so $\|X\|_1 = 2\sqrt{1 - F^2}$ and

$$\Delta \leq 2\sqrt{1 - F^2}. \quad (3)$$

Step 2 (energetics). The survival amplitude is $\chi(T) = \sum_n p_n e^{-iE_n T}$ and $F \geq \text{Re} \chi(T)$. By the tangent-line lemma below, $1 - \cos x \leq Kx$ for all $x \geq 0$, so

$$1 - F \leq 1 - \text{Re} \chi(T) = \sum_n p_n (1 - \cos E_n T) \leq K T \langle H \rangle. \quad (4)$$

Combine. $1 - F^2 = (1 - F)(1 + F) \leq 2(1 - F) \leq 2KT \langle H \rangle$. Inserting into (3), $\Delta^2 \leq 4(1 - F^2) \leq 8KT \langle H \rangle$, i.e. $T \langle H \rangle \geq \Delta^2 / (8K) = C_* \Delta^2$ (with $\sigma_A = 1$). Restoring σ_A gives the claim. Retaining the exact geometry $1 - F \geq 1 - \sqrt{1 - \delta^2}$ ($\delta = \Delta / 2\sigma_A$) before the step $1 - F^2 \leq 2(1 - F)$ yields the sharper non-asymptotic form $T \langle H \rangle \geq (1 - \sqrt{1 - \delta^2}) / K$, whose leading term is $C_* \Delta^2 / \sigma_A^2$ and which is strictly stronger for every finite δ (by a factor approaching 2 as $\delta \rightarrow 1$). $\square$

The energetics (Step 2)—ground-referencing H , the survival amplitude $\chi(T)$, and the eigenbasis cosine estimate—follows the statistical-distance derivation of the Margolus–Levitin bound by Jones and Kok [11] (corrected by Zwierz [12]) and the survival-amplitude inequality of Bhattacharyya [13]; the new step is to place a bounded observable on the left via Hölder (Step 1), converting the state-distance bound into one on $\Delta \langle A \rangle$. Since $\chi(T)$ contains only populated levels, the bound also holds with $\langle H \rangle$ referenced to the lowest *populated* energy $E_{\min}^{\text{supp}} \geq E_0$, a state-dependent strengthening.

Lemma 1 (Tangent line). *$K := \sup_{x>0} (1 - \cos x)/x$ is attained at the unique stationary point x_* of $(1 - \cos x)/x$ in $(0, 2\pi)$ —equivalently, the smallest positive root of $x \sin x = 1 - \cos x \Leftrightarrow \tan(x/2) = x$ —and $K = \sin x_*$. The line $y = Kx$ is tangent to $1 - \cos x$ at x_* and lies above it for all $x \geq 0$, with equality only at $x = 0$ and $x = x_*$.*

This tangent-line constant is the orthogonality ($q \rightarrow 0$) member of the generalized Margolus–Levitin construction of Giovannetti, Lloyd, and Maccone [14]; the same $K \approx 0.7246$ underlies the operator-flow bound of Ref. [8]. Applying the theorem to $E_{\max} I - H \succeq 0$ (which shares F , σ_A , and Δ) gives the ceiling-referenced companion $T(E_{\max} - \langle H \rangle) \geq C_* \Delta^2 / \sigma_A^2$ with the *same* constant; combined, $T \geq C_* \Delta^2 / [\sigma_A^2 \min(\langle H \rangle - E_0, E_{\max} - \langle H \rangle)]$ —the observable analog of the bounded-spectrum (dual)

	resource = variance ΔH (Mandelstam–Tamm)	resource = mean energy $\langle H \rangle - E_0$ (Margolus–Levitin)
state, $1 - F$	$L \leq \Delta H T$	$1 - F \leq K T (\langle H \rangle - E_0)$ (linear)
observable, $\Delta \langle A \rangle$	$T \geq \Delta / (2 \Delta H \sigma_A)$ (linear)	$T (\langle H \rangle - E_0) \geq C_* \Delta^2 / \sigma_A^2$ (quadratic; this work)

TABLE I. The speed-limit dichotomy: *what moves* (a state or an observable) $\times$ *which energy functional* powers the bound (variance or mean energy). Three corners were known; the lower-right (mean energy $\times$ observable) corner was empty and is the *only* quadratic one.

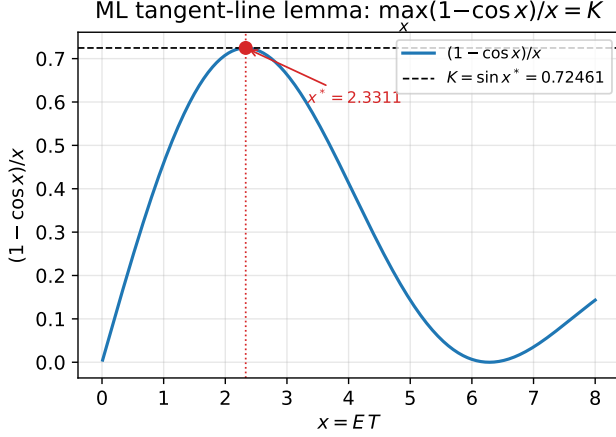


FIG. 1. The tangent-line lemma. $(1 - \cos x)/x$ attains its supremum $K = \sin x_* = 0.724611\dots$ at $x_* = 2.33112\dots$ (root of $\tan(x/2) = x$), which fixes the sharp constant $C_* = 1/(8K)$.

Margolus–Levitin bound [15], whose floor is weakest at mid-spectrum and does not vanish as $\langle H \rangle$ grows. The proof uses only the pointwise tangent inequality and a finite mean energy, so it extends to a separable infinite-dimensional Hilbert space with bounded A and a spectral measure of finite first moment.

Figure 1 shows the lemma. All inequalities are elementary and exact; we verified the chain numerically over 10^5 random configurations in dimensions 2–7 with zero violations (Sec. VIII).

V. EXACT CONSTANT, TRADE-OFF CURVE, AND SATURATION

The constant admits several equivalent closed forms,

$$C_* = \frac{1}{8K} = \frac{1}{8 \sin x_*} = \frac{x_*}{8(1 - \cos x_*)} = \frac{1 + x_*^2}{16 x_*}, \quad (5)$$

all equal to $0.172506267461\dots$ and agreeing to 52 digits (`mpmath`). It is dimension-independent and *sharp*: it is approached, though not attained, by the near-ground two-level family $H = \text{diag}(0, E)$, $A = \sigma_x$ ($\sigma_A = 1$), $|\psi\rangle = \cos \theta |0\rangle + \sin \theta e^{i\beta} |1\rangle$, in the limit $\theta \rightarrow 0$ with $ET = x_*$ and the *optimal relative phase* $\beta = x_*/2 - \pi/2$ (which

starts $\langle A(t) \rangle$ at maximum slope rather than at a turning point). There all three inequalities saturate simultaneously. A dedicated global optimizer over (E_n, A, ψ) in $d = 2\text{--}10$ confirms the floor: the smallest ratio found is $0.172506558 \geq C_*$, approached from above as $\theta \rightarrow 0$, and no configuration dips below (Fig. 5). Phase restriction ($\beta = 0$, real amplitudes) gives instead the strictly larger $C_{\beta=0} = \frac{1}{16 \phi_* \sin^2 \phi_*} = \frac{\phi_*}{4(1 - \cos \phi_*)^2} = 0.185551\dots$ (+7.6%), with $\phi_* = 2.78650\dots$ the smallest positive root of $1 - \cos x = 2x \sin x$, saturated at the turning-point configuration; the optimal relative phase recovers the full factor (Fig. 3).

The complete trade-off curve. The constant C_* is the $\delta \rightarrow 0$ slope of a sharp curve. With budget $P = T(\langle H \rangle - E_0)$ and normalized swing $\delta = \Delta/2\sigma_A \in [0, 1)$, the minimum budget that produces swing δ —over all (H, A, ρ, d, T) —is $P_*(\delta) = \min\{T(\langle H \rangle - E_0) : \Delta/2\sigma_A = \delta\}$, attained on a two-level Bohr-active subspace and given parametrically in the Bohr phase $\tau = ET \in [x_*, \pi)$ by

$$\begin{aligned} \sin^2 \theta &= \frac{\tau - \tan(\tau/2)}{\tau - 2 \tan(\tau/2)}, & P_* &= \tau \sin^2 \theta, \\ \delta &= \sin 2\theta \sin(\tau/2), & \tau &= ET \in [x_*, \pi). \end{aligned} \quad (6)$$

It interpolates from $P_*(\delta) = 4C_*\delta^2[1 + O(\delta^2)]$ as $\delta \rightarrow 0$ (recovering the constant) to $P_* \rightarrow \pi/2$ as $\delta \rightarrow 1$ (Fig. 2), lying strictly above the elementary bound $(1 - \sqrt{1 - \delta^2})/K$ of Sec. IV—by 1.3% at $\delta = 0.5$, 4.2% at $\delta = 0.8$, 8.2% at $\delta = 0.95$, but under 1% in the near-ground window where the applications operate. The curve is the *observable image* of the Giovannetti–Lloyd–Maccone state trade-off: $\Delta = 2\sigma_A \sqrt{1 - F^2}$ is saturable (align the $\pm\sigma_A$ eigenvectors of A with the rank-two $\rho_T - \rho_0$), so the maximal swing at fixed budget is $2\sigma_A \sqrt{1 - F_{\min}^2}$ with $F_{\min}$ the minimal survival fidelity at fixed mean energy [14], whose two-level optimizer [16] carries over; a multi-start search over $d = 2\text{--}6$ finds no state below $P_*(\delta)$ at any δ . This is the observable counterpart—tight at every δ , not only in the constant—of the exact arbitrary-fidelity Margolus–Levitin curve for states.

Bandwidth-resolved constant. If $\text{spec}(H) \subset [E_0, E_0 + B]$, the tangent-line supremum runs only over $x \in (0, BT]$, so $K \rightarrow K(L) = \sup_{0 < x \leq L} (1 - \cos x)/x$ with $L = BT$. Since $(1 - \cos x)/x$ increases on $(0, x_*)$, $K(L) = (1 - \cos L)/L$ for $L < x_*$ and $K(L) = K$ otherwise, giving

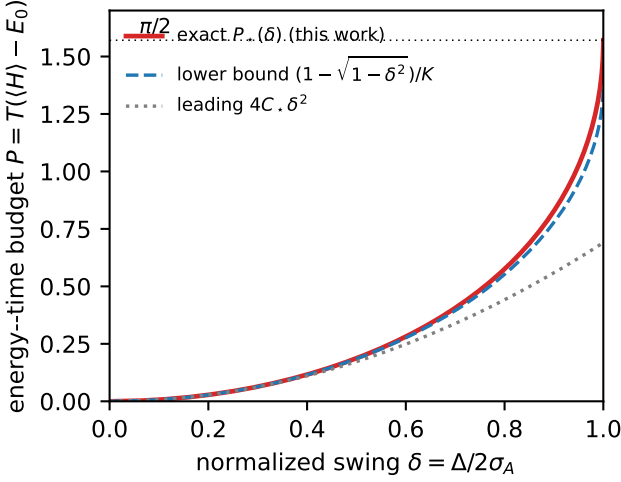


FIG. 2. The exact observable trade-off curve $P_*(\delta)$, Eq. (6): the minimum energy-time budget $P = T(\langle H \rangle - E_0)$ for a normalized swing $\delta = \Delta/2\sigma_A$, tight at every δ . Its slope as $\delta \rightarrow 0$ is set by C_* and its $\delta \rightarrow 1$ limit is $\pi/2$. It lies above the elementary lower bound $(1 - \sqrt{1 - \delta^2})/K$ and its leading term $4C_*\delta^2$.

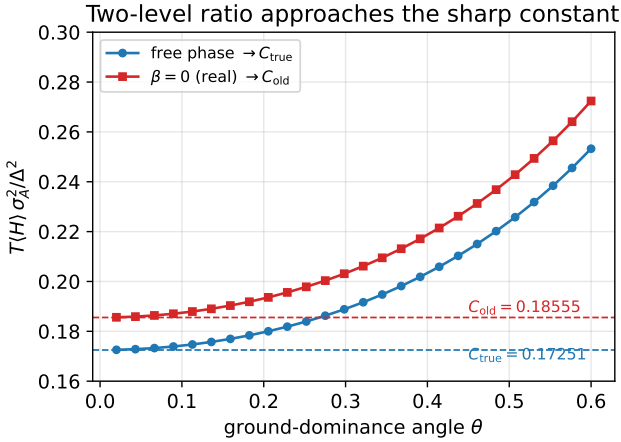


FIG. 3. The two-level ratio $T\langle H \rangle \sigma_A^2 / \Delta^2$ versus the ground-dominance angle θ , approaching the sharp constant C_* (free relative phase) and the phase-restricted suboptimum $C_{\beta=0} = 0.185551$, both from above as $\theta \rightarrow 0$.

the spectrally sharp constant

$$C_*(L) = \frac{1}{8K(L)}, \quad T(\langle H \rangle - E_0) \geq C_*(BT) \frac{\Delta^2}{\sigma_A^2}, \quad (7)$$

which recovers C_* for $L \geq x_*$ and is strictly stronger for narrow-band, short-time operation ($1.58\times$ at $L = 1$, $2.96\times$ at $L = 0.5$), saturated by a two-level gap at the band edge—sharpening the floor for precisely the low-bandwidth devices (e.g. minimal clocks) where the bound is most relevant.

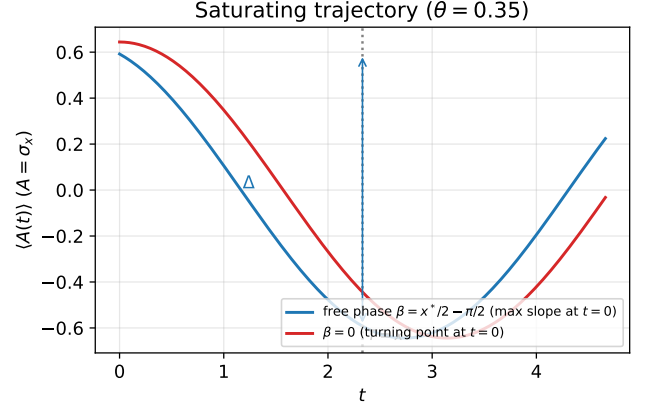


FIG. 4. Saturating trajectory $\langle A(t) \rangle$ ($A = \sigma_x$). With the optimal phase the expectation starts at maximum slope (and reaches its excursion Δ at $T = x_*/E$); the real-amplitude ($\beta = 0$) case starts at a turning point and is suboptimal.

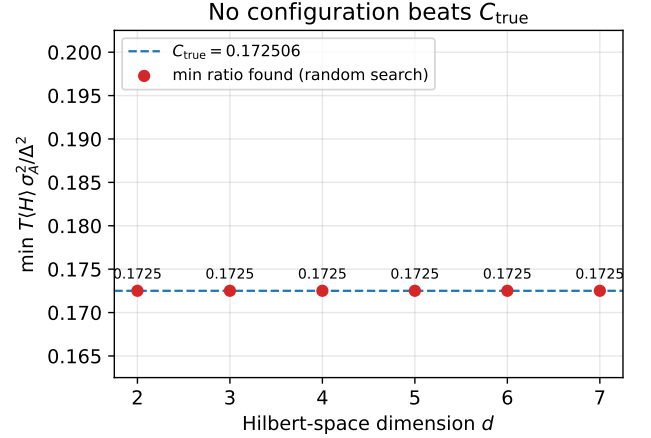


FIG. 5. Global-optimizer minimum of $T\langle H \rangle \sigma_A^2 / \Delta^2$ per Hilbert-space dimension d . No configuration in $d = 2-7$ beats C_* ; each dimension approaches it from above.

VI. MIXED STATES AND NORMALIZATION

The bound holds for an arbitrary density matrix ρ_0 with the same constant C_* , with $\rho_T = U\rho_0U^\dagger$, $U = e^{-iHT}$, $\Delta = |\text{Tr}[A(\rho_T - \rho_0)]|$, $\langle H \rangle = \text{Tr}[\rho_0 H]$. The geometric step generalizes via Hölder to $\Delta \leq 2\sigma_A D_{\text{tr}}(\rho_0, \rho_T)$, with D_{tr} the trace distance. The key is to avoid an Uhlmann-fidelity detour (which loses a factor of two) by using *joint convexity* of the trace distance on the spectral decomposition $\rho_0 = \sum_k p_k |k\rangle\langle k|$: since the *same* U evolves every eigenvector, $\rho_T = \sum_k p_k U |k\rangle\langle k| U^\dagger$ is a convex combination with the *identical* weights p_k , so

$$D_{\text{tr}}(\rho_0, \rho_T) \leq \sum_k p_k \sqrt{1 - F_k^2}, \quad F_k = |\langle k|U|k \rangle|. \quad (8)$$

Per-eigenstate ML gives $1 - F_k \leq KT e_k$ with $e_k = \langle k|H|k \rangle \geq 0$, and Jensen's inequality (concavity of $\sqrt{\cdot}$, $\sum_k p_k e_k = \langle H \rangle$) yields $\sum_k p_k \sqrt{2KT e_k} \leq \sqrt{2KT \langle H \rangle}$. Combining, $\Delta \leq 2\sigma_A \sqrt{2KT \langle H \rangle}$, hence $T \langle H \rangle \geq C_* \Delta^2 / \sigma_A^2$. The same constant follows independently by purification ($H \otimes I$, $A \otimes I$ preserve $\langle H \rangle$, σ_A and Δ exactly, so the pure bound descends). Each inequality is strict away from the pure $\theta \rightarrow 0$ limit, so every state with $\Delta > 0$ lies strictly above C_* ; yet C_* remains the tight infimum for mixed states as well (no pure/mixed gap), approached by mixing the saturating family with a vanishing ground-state weight, consistent with the saturation analysis of Ref. [17].

Among candidate normalizations, the spectral spread $\sigma_A = (\lambda_{\max} - \lambda_{\min})/2$ is canonical: it is the Chebyshev radius $\min_c \|A - cI\|_{\text{op}}$, which makes the Hölder step tight; it is shift-invariant and state-independent. The operator norm $\|A\|_{\text{op}} \geq \sigma_A$ gives a weaker valid bound, while the state variance $\Delta A_\rho \leq \sigma_A$ (Popoviciu) is the variance scale of the MT family and is not delivered by this proof.

Several observables at once. Because the bound depends on the state only through $X = \rho_T - \rho_0$, it constrains a whole family $\{A_i\}$ jointly: for any real combination $A_u = \sum_i u_i A_i$ with swing $\Delta_u = |\text{Tr}[A_u X]|$ and Chebyshev radius $\|A_u\|_{\text{Cheb}}$, Step 1 gives $\Delta_u \leq \|A_u\|_{\text{Cheb}} 2\sqrt{1 - F^2}$, hence

$$T(\langle H \rangle - E_0) \geq C_* \sup_u \frac{\Delta_u^2}{\|A_u\|_{\text{Cheb}}^2}, \quad (9)$$

a mean-energy bound set by the operator-norm (Chebyshev) *dual* geometry, in contrast to the Fisher-metric multiparameter variance limits [7, 18]. For two orthonormal observables (e.g. σ_x, σ_y) it reads $T(\langle H \rangle - E_0) \geq C_*(\Delta_1^2 + \Delta_2^2)/\sigma_A^2$, twice the single-observable floor when $\Delta_1 = \Delta_2$.

VII. STRUCTURE: LINEAR RECOVERY AND STRUCTURED OBSERVABLES

The quadratic law is a property of *generic* observables, whose expectation starts at maximum slope. It degrades to a *linear* law when the initial state is an eigenvector of A (a sufficient condition).

Theorem 3 (Linear recovery). *If $A|\psi_0\rangle = \lambda|\psi_0\rangle$, then with $l = (\lambda - \text{center}(\text{spec } A))/\sigma_A \in [-1, 1]$,*

$$T(\langle H \rangle - E_0) \geq \frac{1}{2K(1 + |l|)} \frac{\Delta}{\sigma_A}, \quad (10)$$

sharpest at an extreme eigenvalue ($|l| = 1$): $T(\langle H \rangle - E_0) \geq \Delta/(4K) = 2C_ \Delta/\sigma_A$.*

Proof. Center A , $\sigma_A = 1$; set $B = A - \lambda I$, so $B|\psi_0\rangle = 0$. Writing $|\psi_T\rangle = e^{i\alpha} F|\psi_0\rangle + \sqrt{1 - F^2}|\phi\rangle$ with $|\phi\rangle \perp |\psi_0\rangle$, the cross terms vanish ($B|\psi_0\rangle = 0$, B Hermitian), so $\Delta = (1 - F^2)|\langle\phi|B|\phi\rangle| \leq (1 + |l|)(1 - F^2)$ —linear in the

fidelity deficit. With $1 - F^2 \leq 2(1 - F) \leq 2KT\langle H \rangle$ the claim follows. $\square$

Thus a fidelity-aligned observable such as the projector $|\psi_0\rangle\langle\psi_0|$ literally reduces the bound to the linear ML *state* bound (then $\langle A(t) \rangle = F^2$, $\Delta = 1 - F^2$), recovering the upper-right corner of the dichotomy. We emphasize that a turning point alone (zero initial slope without $|\psi_0\rangle$ being an eigenvector) is *not* sufficient: e.g. $|\psi_0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, $A = \text{diag}(1, -1, 0)$, H with real H_{01} has zero initial slope yet no positive linear constant (r_{lin} falls to $\approx 0.03 \ll 1/4K$).

For structured observables the constant does not improve generically: a rank-one ground coupling $A = |0\rangle\langle v| + |v\rangle\langle 0|$ still gives $\inf = C_*$ (the saturating family is itself of this form), and an energy-banded ground-touching A likewise. Restricting to real couplings and amplitudes raises the constant to $C_{\beta=0} = 0.185551$. A ground-decoupled observable ($A_{0n} = 0$) conserves the excited population and makes the time bound vacuous; a static energy floor $\langle H \rangle - E_0 \geq (\Delta_{\text{gap}}/2)\Delta/\sigma_A$ governs instead. The bound is informative for observables of finite spectral spread (spins, parity, projectors, qubit registers); for unbounded observables ($\sigma_A = \infty$, e.g. position or particle number) it is vacuous, and Mandelstam–Tamm, which uses the finite state variance, is operative.

VIII. NUMERICAL VERIFICATION

All numerics are seeded and reproducible (`numerics/`; consolidated in `results.json`). The named constant is pinned to 52 digits with `mpmath`. The no-go exponent is confirmed by the log-log slope of $P_{\min}(\Delta)$ along the saturating family (2.0). The pure-state bound survives 10^5 random trials in $d = 2\text{--}7$ (worst-case slacks ≥ 0 to machine precision) and a dedicated multi-start optimizer (Fig. 5); the apparent sub- C_* minima are floating-point catastrophic cancellation at excited population $\sim 10^{-14}$ and disappear at 60-digit precision. The mixed-state bound is confirmed by a violation search over random density matrices (minimum sampled ratio $0.176 \geq C_*$) and by a direct audit of the convexity chain; the infimum over mixed states is again C_* , approached as the state purifies onto the saturating family. The linear-recovery constant $1/4K$ and the structured constants C_* , $C_{\beta=0}$ are reproduced to six digits. The exact trade-off curve $P_*(\delta)$ is confirmed as a strict lower boundary by a multi-start search in $d = 2\text{--}6$ (no state dips below it at any δ); the bandwidth-resolved bound $C_*(BT)$ by a violation search over spectrally bounded H (0 violations in 1.5×10^5 trials); and the two-observable bound saturates the $2\times$ gain at $\Delta_1 = \Delta_2$.

IX. APPLICATIONS AND DISCUSSION

The most important consequence is structural: Eq. (1) is the first mean-energy, quadratic-in- Δ member of the observable-QSL family, the variance/Fisher members being *linear* in Δ [5–7]. Being quadratic, it is most constraining exactly where the linear bounds are weakest—for *large* fractional changes Δ/σ_A —so the two families are complementary, not competing. Observable QSLs are an active, growing area, including recent experiments [9].

a. A regime where the bound binds. There is an identifiable window in which Eq. (1) is the operative observable limit. For the near-ground qubit $H = \frac{\omega}{2}\sigma_z$, $A = \sigma_x$, $|\psi_0\rangle = \sqrt{1-\varepsilon}|g\rangle + \sqrt{\varepsilon}|e\rangle$ (excited population ε , ground state $|g\rangle$) evaluated at the half-period $T_\star = \pi/\omega$, the mean-energy floor exceeds the Mandelstam–Tamm observable floor $\Delta/(2\Delta H\sigma_A)$ by the exact factor $8C_\star(1-\varepsilon)$, which is > 1 for ground dominance $\varepsilon < 0.275$. There—small excited population, large fractional swing—the quadratic mean-energy bound, not the variance bound, sets the limit. Read in reverse, an observed swing Δ over time T *certifies* a mean energy $\langle H \rangle - E_0 \geq C_\star\Delta^2/(\sigma_A^2 T)$ from a single bounded-observable expectation, with no state-overlap measurement. With Δ estimated from finite samples (standard error s), error propagation gives the one-sided certificate $\langle H \rangle - E_0 \geq C_\star[\max(0, \hat{\Delta} - z_\alpha s)]^2/(\sigma_A^2 T)$ at confidence $1 - \alpha$ —a model-independent energy floor measurable on the platforms now realizing observable speed limits [9], in the regime above where it binds.

b. Autonomous quantum clocks (primary). An autonomous clock is by definition driven by a fixed, time-independent Hamiltonian [19, 20], so the hypotheses of Eq. (1) hold exactly. Taking A to be the clock’s bounded pointer (hand) observable and Δ a resolvable advance, the bound becomes a coherent *resolution floor*,

$$T_{\text{res}} \geq \hbar C_\star \frac{\Delta^2}{(\langle H \rangle - E_0) \sigma_A^2}, \quad (11)$$

i.e. the mean energy stored in the clockwork lower-bounds the time to move the hand by a resolvable amount. A precessing pointer ($H = \frac{\omega}{2}\sigma_z$, $A = \sigma_x$) obeys it, with the bound tightest for the smallest, lowest-energy “minimal” clock; saturation is a near-ground limit, and at its natural (equatorial) operating point the pointer is an A -turning point—the linear regime of Sec. VII—so it illustrates rather than saturates the quadratic floor. This complements—rather than subsumes—the established clock limits, which use *different* resources: dissipated entropy per tick ($N \lesssim \Delta S_{\text{tick}}/2k_B$) [19, 21] and the thermalization rate ($N \leq \Gamma^2/\nu^2$) [22], both for open dissipative clocks. Mean energy alone does not *upper-bound* clock or computational speed [23]; ours is a complementary lower bound at fixed pointer contrast σ_A .

c. Autonomous quantum batteries (secondary). For an autonomous (time-independent generator) battery with $A = H_B$ the stored energy, $\Delta = W$ the energy deposited, and σ_B the battery’s spectral spread,

Eq. (1) gives a minimum charging time $T_{\text{min}} = \hbar C_\star W^2/[(\langle H_{\text{tot}} \rangle - E_0)\sigma_B^2]$, quadratic in W . This is a *complementary* mean-energy floor: mainstream charging is driven by time-dependent control and its operative power bounds are variance/Fisher based [24–26] (the García-Pintos bound has a published correction [27]), and the reported quantum advantage stems from super-extensive *variance*, not mean energy. The same algebra rearranges into an energetic-cost floor on changing a logical observable in any always-on (autonomous) architecture; it is, however, loose for typical fast gates, and standard time-dependent gate pulses fall outside the hypotheses. Because σ_B tracks the battery bandwidth, this floor falls as $1/N$ for an N -cell battery and sits a fixed factor below the Margolus–Levitin charging time even at $N = 1$, and it bounds the deposited energy W rather than the ergotropy [28]; we therefore offer it as an illustrative remark, not a binding constraint.

d. What the bound does not constrain. Honesty about scope sharpens the claim. (i) Standard, time-dependently driven quantum gates violate the time-independence hypothesis—essential here, since the Margolus–Levitin bound provably does not extend to closed systems with time-dependent generators [16, 29]. (ii) For extensive many-body systems the total mean energy grows like the system size, so the bound is weak for *local* observables; there the operative limit is locality-based (Lieb–Robinson) [30]. (iii) In quantum metrology/sensing the operative speed limit is the variance (MT) one and practical readout times are set by signal-to-noise and decoherence far above any mean-energy floor [31]; the bound nonetheless lower-bounds the mean-energy *cost* of generating a given signal swing, complementing the asymmetry/weak-value observable speed limit of Budiyono *et al.* [32]. The bound is therefore a statement about energy-constrained, few-body or near-ground dynamics.

e. Outlook. Natural extensions are open-system (Lindbladian) and time-dependent- H generalizations—where much of the driven-application landscape lives—and a sharp constant for ergotropy rather than stored energy. Within the present setting, the constant sharpens under a spectral-bandwidth restriction ($K \rightarrow \sup_{0 < x \leq BT} (1 - \cos x)/x$) and generalizes to higher energy moments [33], and the bound assembles with the observable Mandelstam–Tamm limit into a unified envelope $T \geq \max[\Delta/(2\Delta H\sigma_A), C_\star\Delta^2/((\langle H \rangle - E_0)\sigma_A^2)]$ in the spirit of Levitin–Toffoli [3] and the bounded-spectrum treatment [15].

X. CONCLUSION

We have established the missing mean-energy speed limit for observables: a no-go against any linear law and a sharp quadratic bound $T(\langle H \rangle - E_0) \geq C_\star\Delta^2/\sigma_A^2$ with a closed-form, dimension-independent constant, valid for pure and mixed states, tight, and degrad-

ing to a linear law for eigenvector observables. Beyond the constant we gave the complete, tight trade-off curve $P_*(\delta)$ (the observable analog of the arbitrary-fidelity Margolus–Levitin curve), its bandwidth-resolved and multi-observable refinements, and its backward read-

ing as a model-independent mean-energy certificate. Together these make the (mean-energy $\times$ observable) corner of the speed-limit dichotomy a complete and operationally grounded one; its cleanest physical home is the thermodynamics of timekeeping.

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